

Coulomb's Law

In our previous observation (experiments) we found that the electric force increases for objects with more charge and decreases as charged objects are moved further apart. Now we are interested to quantify this observation.

The force law that describes this behavior is known as Coulomb's law.

Coulomb announced in 1785 that the electric force obeys an inverse-square law analogous to Newton's law of gravity.

The law can be stated as follows

1. If two charged particles having charges q_1 and q_2 are a distance r apart, the particles exert forces on each other of magnitude.

$$F_{1 \text{ on } 2} = F_{2 \text{ on } 1} = \frac{k |q_1| |q_2|}{r^2} \quad \text{--- (1)}$$

where k is called the electrostatic constant.

These forces are an action/reaction pair, equal in magnitude and opposite in direction.

2. The forces are directed along the line joining the two particles. The forces are repulsive for two like charges and attractive for two opposite charges.

Electrostatic force:

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r} \quad (\text{Coulomb's law})$$

Newton's Gravitational force

$$\vec{F} = G \frac{m_1 m_2}{r^2} \hat{r} \quad (\text{Newton's law})$$

Two types of forces are widely different, both equations describe inverse square laws ($\frac{1}{r^2}$ dependence) that involve a product of a property of the interacting particles, — the charge in one case and the mass in the other.

— The laws differ in that gravitational forces are always attractive but electrostatic forces may be either attractive or repulsive, depending on the signs of the charges.

→ Reasons, one type of mass but two types of charge.

Unit of charge

Coulomb had no unit of charge, so he was unable to determine a value for k , whose numerical value depends upon the units of both charge and distance.

The SI unit of charge, the Coulomb (C), is derived from the SI unit of current. ($i = \frac{dq}{dt}$)

Unit Charge $e = 1.60 \times 10^{-19} \text{ C}$.

State another way: 1C is the net charge on roughly 6.25×10^{18} protons.

Once the unit of charge is established, torsion balance experiments — can be used to measure the electrostatic constants k . The SI units (17)

$$k = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2$$

Surprisingly, we will find that Coulomb's law is not explicitly used in much of the theory of electricity. While it is the basic force law, most of our future discussion and calculations will be of things called fields and potentials.

If we rewrite Coulomb's law in a somewhat more complicated way, let's define a constant, called the permittivity constant ϵ_0 as,

$$\epsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12} \text{ C}^2/\text{N m}^2$$

Rewriting Coulomb's law in terms of ϵ_0 gives us

$$F = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$$

Multiple forces:

$$\vec{F}_{\text{net}} = \vec{F}_{1onj} + \vec{F}_{2onj} + \vec{F}_{3onj} + \dots +$$